# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 8 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** The sum of the consecutive triangular numbers $T_6$ and $T_7$ is: 1

**(a)** 56 **(b)** 36 **(c)** 49 **(d)** 50

**2.** Which of the following equals $4.07 \times 10^{-3}$? 1

**(a)** 4070 **(b)** 0.00407 **(c)** 0.0407 **(d)** 0.000407

**3.** A Babylonian numeral has 2 group(s) of 60 and 15 units. Its value is: 1

**(a)** 27 **(b)** 105 **(c)** 17 **(d)** 135

**4.** The diagonals of a rectangle ABCD meet at O and $\angle AOB = 80^\circ$. Then $\angle OAB$ is: 1

**(a)** $40^\circ$ **(b)** $50^\circ$ **(c)** $100^\circ$ **(d)** $80^\circ$

**5.** The sum of the three consecutive numbers 12, 13 and 14 is: 1

**(a)** 37 **(b)** 42 **(c)** 36 **(d)** 39

**6.** For m = 4, the Pythagorean triplet $(2m,\ m^2-1,\ m^2+1)$ is: 1

**(a)** 8, 14, 18 **(b)** 8, 15, 17 **(c)** 8, 17, 18 **(d)** 4, 15, 17

**7.** $\frac{1}{64}$ expressed as a power of 2 is: 1

**(a)** $2^{-5}$ **(b)** $2^{-7}$ **(c)** $2^{6}$ **(d)** $2^{-6}$

**8.** A Maya numeral has 3 in the 20s place and 12 in the units place. Its value is: 1

**(a)** 15 **(b)** 72 **(c)** 42 **(d)** 48

**9.** In a kite ABCD with AB = AD and CB = CD, $\angle B = \angle D = 90^\circ$ and $\angle A = 70^\circ$. Then $\angle C$ is: 1

**(a)** $200^\circ$ **(b)** $130^\circ$ **(c)** $110^\circ$ **(d)** $90^\circ$

**10.** The sum of the five consecutive numbers starting from 14 is: 1

**(a)** 70 **(b)** 85 **(c)** 75 **(d)** 80

**11.** The sum of the cubes of the first 4 natural numbers is: 1

**(a)** 100 **(b)** 104 **(c)** 20 **(d)** 10

**12.** In $7^4$, the base is: 1

**(a)** 28 **(b)** 2401 **(c)** 4 **(d)** 7

**13.** The number system we use today is called the: 1

**(a)** Hindu-Arabic system **(b)** Roman system
**(c)** Babylonian system **(d)** Maya system

**14.** In an isosceles trapezium ABCD with AB $\parallel$ DC, $\angle A = 65^\circ$. Then $\angle D$ is: 1

**(a)** $75^\circ$ **(b)** $115^\circ$ **(c)** $65^\circ$ **(d)** $295^\circ$

**15.** Think of 7. Add 5, multiply by 2 and subtract 10. The result is: 1

**(a)** 19 **(b)** 15 **(c)** 14 **(d)** 9

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** Find the Pythagorean triplet whose smallest member is 6. 2

**17.** Write $\left(\frac{2}{3}\right)^{-3}$ in simplest form. 2

**18.** Write 29 in the binary system. 2

**19.** In a parallelogram ABCD, $\angle A = (3x + 10)^\circ$ and $\angle B = (2x - 10)^\circ$. Find $x$ and $\angle A$. 2

**20.** Apply one step of the Kaprekar routine to 5632. 2

## SECTION — C (Short Answer Type) 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** Find XLIV + LXVII and write the answer in Roman numerals. 3

**22.** ABCD is a rhombus whose diagonals meet at O. If $\angle BAD = 64^\circ$, find $\angle OAB$, $\angle ABC$ and $\angle OBA$. 3

**23.** Apply the Kaprekar routine to 3521 repeatedly until you reach 6174. Write each step. 3

**24.** Find $\sqrt{1156}$ by prime factorisation. 3

OR

Express the following in standard form: 3
(i) 5900000
(ii) 0.00074
(iii) 80000.

**25.** Write $2431_5$ in expanded form and find its decimal value. 3

OR

The diagonals of a rectangle ABCD meet at O and $\angle AOB = 70^\circ$. Find $\angle OAB$, $\angle OAD$ and $\angle AOD$. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** Replace the stars so that the number 4*3* is divisible by both 5 and 9. Find all possibilities. 5

OR

Find $\sqrt[3]{13824}$ by prime factorisation and verify the result by cubing. 5

**27.** Write 600000 and 2000 in standard form. Then find their product and quotient in standard form. 5

OR

A shopkeeper has weights of 1, 2, 4, 8, 16 and 32 kg. Show how to weigh exactly 13 kg using these weights, each at most once. Explain the idea using binary numbers. 5

**END OF QUESTION PAPER** **Total: 50 Marks**

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